



MATH110: Descriptive Statistics

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Box-and-Whisker Plot

1. Find the five-number summary (Min, Q1, Median, Q3, Max) for the data set below, then construct a box-and-whisker plot.

{14, 16, 21, 24, 25, 28, 38, 41, 50, 56, 57, 60}

Construct your box-and-whisker plot here



Answer: _____

2. Find the five-number summary (Min, Q1, Median, Q3, Max) for the data set below, then construct a box-and-whisker plot.

{10, 14, 19, 21, 30, 36, 41, 49, 56, 59}

Construct your box-and-whisker plot here



Answer: _____



3. Find the five-number summary (Min, Q1, Median, Q3, Max) for the data set below, then construct a box-and-whisker plot.

{10, 13, 25, 28, 30, 32, 36, 39, 41, 44, 56, 59}

Construct your box-and-whisker plot here



Answer: _____

Empirical rule (68-95-99.7)

4. Scores are normally distributed. What percent of scores fall within 3 standard deviation(s) of the mean?

$\mu = 61$, $\sigma = 8$, [37, 85]

Answer: _____

5. Scores are normally distributed. What percent of scores fall within 2 standard deviation(s) of the mean?

$\mu = 79$, $\sigma = 7$, [65, 93]

Answer: _____

6. Scores are normally distributed. What percent of scores fall within 2 standard deviation(s) of the mean?

$\mu = 74$, $\sigma = 11$, [52, 96]

Answer: _____

Quartiles and IQR

7. Find Q1, Q3, and the IQR:

{7, 18, 20, 26, 28, 29, 32, 38}

Answer: _____

8. Find Q1, Q3, and the IQR:

{1, 6, 8, 14, 27, 37}

Answer: _____



9. Find Q1, Q3, and the IQR:

{5, 11, 12, 20, 21, 27, 29, 33}

Answer: _____

Mean of a data set

10. Find the mean of the data set:

{18, 40, 15, 16, 8, 13, 19}

Answer: _____

11. Find the mean of the data set:

{3, 5, 36, 15, 34}

Answer: _____

12. Find the mean of the data set:

{17, 1, 23, 38, 10, 28, 5, 25}

Answer: _____

Median of a data set

13. Find the median of the data set:

{4, 6, 22, 24, 31}

Answer: _____

14. Find the median of the data set:

{6, 16, 17, 37, 38}

Answer: _____

15. Find the median of the data set:

{2, 10, 21, 24, 34}

Answer: _____

Mode of a data set

16. Find the mode:

{2, 4, 6, 11, 11, 11, 17, 18, 19}

Answer: _____



17. Find the mode:

{9, 10, 14, 15, 19, 20, 20}

Answer: _____

18. Find the mode:

{2, 6, 6, 9, 11, 12, 13, 20}

Answer: _____

Outlier detection (1.5xIQR rule)

19. Using the 1.5xIQR rule, determine if the value is an outlier:

$Q_1 = 15$, $Q_3 = 24$, $IQR = 9$, $x = 34$

Answer: _____

20. Using the 1.5xIQR rule, determine if the value is an outlier:

$Q_1 = 11$, $Q_3 = 22$, $IQR = 11$, $x = 37$

Answer: _____

21. Using the 1.5xIQR rule, determine if the value is an outlier:

$Q_1 = 19$, $Q_3 = 27$, $IQR = 8$, $x = 38$

Answer: _____

Range

22. Find the range:

{1, 17, 42, 28, 50}

Answer: _____

23. Find the range:

{23, 33, 13, 5, 49, 4}

Answer: _____

24. Find the range:

{18, 16, 15, 49, 20}

Answer: _____

Sample variance and standard deviation



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Labanan ang agwat.

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25. Find the sample variance and standard deviation (round to 2 decimal places):

{11, 6, 8, 12, 2, 4, 18}

Answer: _____

26. Find the sample variance and standard deviation (round to 2 decimal places):

{16, 8, 19, 13, 20, 12}

Answer: _____

27. Find the sample variance and standard deviation (round to 2 decimal places):

{17, 2, 14, 18}

Answer: _____

Z-score

28. Find the z-score and interpret it:

$x = 80$, $\mu = 65$, $\sigma = 11$

Answer: _____

29. Find the z-score and interpret it:

$x = 96$, $\mu = 81$, $\sigma = 5$

Answer: _____

30. Find the z-score and interpret it:

$x = 89$, $\mu = 70$, $\sigma = 11$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Sample variance and standard deviation, Z-score, Range, Empirical rule (68-95-99.7), Box-and-Whisker Plot, Median of a data set, Quartiles and IQR, Outlier detection (1.5xIQR rule), Mode of a data set, Mean of a data set. All answers verified by independent computation.

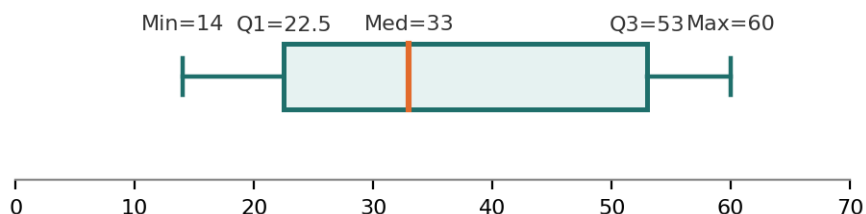
Solutions



Box-and-Whisker Plot

1. Find the five-number summary (Min, Q1, Median, Q3, Max) for the data set below, then construct a box-and-whisker plot.

{14, 16, 21, 24, 25, 28, 38, 41, 50, 56, 57, 60}



→ Sort the data in ascending order: [14, 16, 21, 24, 25, 28, 38, 41, 50, 56, 57, 60].

→ Min = 14, Max = 60.

→ Median = 33.0.

→ Q1 (median of lower half) = 22.5; Q3 (median of upper half) = 53.0.

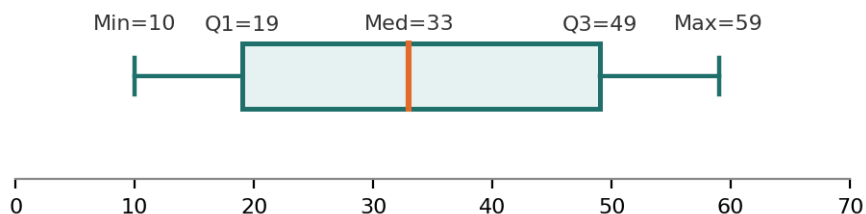
→ IQR = Q3 - Q1 = 30.5.

→ Draw the box from Q1 to Q3 with a line at the median; extend whiskers to Min and Max (see plot).

Answer: Min = 14, Q₁ = 22.5, Med = 33.0, Q₃ = 53.0, Max = 60

2. Find the five-number summary (Min, Q1, Median, Q3, Max) for the data set below, then construct a box-and-whisker plot.

{10, 14, 19, 21, 30, 36, 41, 49, 56, 59}



→ Sort the data in ascending order: [10, 14, 19, 21, 30, 36, 41, 49, 56, 59].

→ Min = 10, Max = 59.

→ Median = 33.0.

→ Q1 (median of lower half) = 19.0; Q3 (median of upper half) = 49.0.

→ IQR = Q3 - Q1 = 30.0.

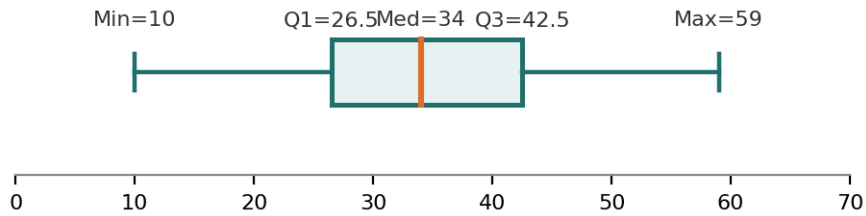
→ Draw the box from Q1 to Q3 with a line at the median; extend whiskers to Min and Max (see plot).

Answer: Min = 10, Q₁ = 19.0, Med = 33.0, Q₃ = 49.0, Max = 59



3. Find the five-number summary (Min, Q1, Median, Q3, Max) for the data set below, then construct a box-and-whisker plot.

{10, 13, 25, 28, 30, 32, 36, 39, 41, 44, 56, 59}



→ Sort the data in ascending order: [10, 13, 25, 28, 30, 32, 36, 39, 41, 44, 56, 59].

→ Min = 10, Max = 59.

→ Median = 34.0.

→ Q1 (median of lower half) = 26.5; Q3 (median of upper half) = 42.5.

→ IQR = Q3 - Q1 = 16.0.

→ Draw the box from Q1 to Q3 with a line at the median; extend whiskers to Min and Max (see plot).

Answer: Min = 10, Q₁ = 26.5, Med = 34.0, Q₃ = 42.5, Max = 59



Empirical rule (68-95-99.7)

4. Scores are normally distributed. What percent of scores fall within 3 standard deviation(s) of the mean?

$$\mu = 61, \quad \sigma = 8, \quad [37, 85]$$

→ Lower bound: $61 - 3(8) = 37$.

→ Upper bound: $61 + 3(8) = 85$.

→ Empirical rule: {answer}% of data falls within {k} standard deviation(s) of the mean.

Answer: $37 = \mu - 3\sigma, \quad 85 = \mu + 3\sigma \Rightarrow 99.7\%$

5. Scores are normally distributed. What percent of scores fall within 2 standard deviation(s) of the mean?

$$\mu = 79, \quad \sigma = 7, \quad [65, 93]$$

→ Lower bound: $79 - 2(7) = 65$.

→ Upper bound: $79 + 2(7) = 93$.

→ Empirical rule: {answer}% of data falls within {k} standard deviation(s) of the mean.

Answer: $65 = \mu - 2\sigma, \quad 93 = \mu + 2\sigma \Rightarrow 95\%$

6. Scores are normally distributed. What percent of scores fall within 2 standard deviation(s) of the mean?

$$\mu = 74, \quad \sigma = 11, \quad [52, 96]$$

→ Lower bound: $74 - 2(11) = 52$.

→ Upper bound: $74 + 2(11) = 96$.

→ Empirical rule: {answer}% of data falls within {k} standard deviation(s) of the mean.

Answer: $52 = \mu - 2\sigma, \quad 96 = \mu + 2\sigma \Rightarrow 95\%$



Quartiles and IQR

7. Find Q1, Q3, and the IQR:

{7, 18, 20, 26, 28, 29, 32, 38}

→ Sort: [7, 18, 20, 26, 28, 29, 32, 38]. $n = 8$.

→ Lower half: [7, 18, 20, 26] → $Q_1 = 19.0$.

→ Upper half: [28, 29, 32, 38] → $Q_3 = 30.5$.

→ $IQR = Q_3 - Q_1 = 30.5 - 19.0 = 11.5$.

Answer: $Q_1 = 19.0$, $Q_3 = 30.5$, $IQR = 11.5$

8. Find Q1, Q3, and the IQR:

{1, 6, 8, 14, 27, 37}

→ Sort: [1, 6, 8, 14, 27, 37]. $n = 6$.

→ Lower half: [1, 6, 8] → $Q_1 = 6.0$.

→ Upper half: [14, 27, 37] → $Q_3 = 27.0$.

→ $IQR = Q_3 - Q_1 = 27.0 - 6.0 = 21.0$.

Answer: $Q_1 = 6.0$, $Q_3 = 27.0$, $IQR = 21.0$

9. Find Q1, Q3, and the IQR:

{5, 11, 12, 20, 21, 27, 29, 33}

→ Sort: [5, 11, 12, 20, 21, 27, 29, 33]. $n = 8$.

→ Lower half: [5, 11, 12, 20] → $Q_1 = 11.5$.

→ Upper half: [21, 27, 29, 33] → $Q_3 = 28.0$.

→ $IQR = Q_3 - Q_1 = 28.0 - 11.5 = 16.5$.

Answer: $Q_1 = 11.5$, $Q_3 = 28.0$, $IQR = 16.5$



Mean of a data set

10. Find the mean of the data set:

{18, 40, 15, 16, 8, 13, 19}

→ Add all 7 values: 129.

→ Divide by $n = \{n\}$: $\{sum_values\} \div \{n\} = \{answer\}$.

Answer: $\bar{x} = 18.4286$

11. Find the mean of the data set:

{3, 5, 36, 15, 34}

→ Add all 5 values: 93.

→ Divide by $n = \{n\}$: $\{sum_values\} \div \{n\} = \{answer\}$.

Answer: $\bar{x} = 18.6$

12. Find the mean of the data set:

{17, 1, 23, 38, 10, 28, 5, 25}

→ Add all 8 values: 147.

→ Divide by $n = \{n\}$: $\{sum_values\} \div \{n\} = \{answer\}$.

Answer: $\bar{x} = 18.375$



Median of a data set

13. Find the median of the data set:

$\{4, 6, 22, 24, 31\}$

→ Sort the data: $[4, 6, 22, 24, 31]$.

→ $n = 5$ — odd — take the middle value.

→ Median = {answer}.

Answer: Median = 22

14. Find the median of the data set:

$\{6, 16, 17, 37, 38\}$

→ Sort the data: $[6, 16, 17, 37, 38]$.

→ $n = 5$ — odd — take the middle value.

→ Median = {answer}.

Answer: Median = 17

15. Find the median of the data set:

$\{2, 10, 21, 24, 34\}$

→ Sort the data: $[2, 10, 21, 24, 34]$.

→ $n = 5$ — odd — take the middle value.

→ Median = {answer}.

Answer: Median = 21



Mode of a data set

16. Find the mode:

{2, 4, 6, 11, 11, 11, 17, 18, 19}

→ Tally each value's frequency.

→ 11 appears 3 times — more than any other value.

→ Mode = {answer}.

Answer: Mode = 11

17. Find the mode:

{9, 10, 14, 15, 19, 20, 20}

→ Tally each value's frequency.

→ 20 appears 2 times — more than any other value.

→ Mode = {answer}.

Answer: Mode = 20

18. Find the mode:

{2, 6, 6, 9, 11, 12, 13, 20}

→ Tally each value's frequency.

→ 6 appears 2 times — more than any other value.

→ Mode = {answer}.

Answer: Mode = 6



Outlier detection (1.5xIQR rule)

19. Using the 1.5xIQR rule, determine if the value is an outlier:

$$Q_1 = 15, \quad Q_3 = 24, \quad IQR = 9, \quad x = 34$$

$$\rightarrow \text{Upper fence} = Q_3 + 1.5 \times IQR = 24 + 13.5 = 37.5.$$

$$\rightarrow \text{Lower fence} = Q_1 - 1.5 \times IQR = 15 - 13.5 = 1.5.$$

$\rightarrow 34$ is within $[1.5, 37.5]$ $\rightarrow 34$ is NOT an outlier.

Answer: Fences: $[1.5, 37.5]$; 34 is NOT an outlier

20. Using the 1.5xIQR rule, determine if the value is an outlier:

$$Q_1 = 11, \quad Q_3 = 22, \quad IQR = 11, \quad x = 37$$

$$\rightarrow \text{Upper fence} = Q_3 + 1.5 \times IQR = 22 + 16.5 = 38.5.$$

$$\rightarrow \text{Lower fence} = Q_1 - 1.5 \times IQR = 11 - 16.5 = -5.5.$$

$\rightarrow 37$ is within $[-5.5, 38.5]$ $\rightarrow 37$ is NOT an outlier.

Answer: Fences: $[-5.5, 38.5]$; 37 is NOT an outlier

21. Using the 1.5xIQR rule, determine if the value is an outlier:

$$Q_1 = 19, \quad Q_3 = 27, \quad IQR = 8, \quad x = 38$$

$$\rightarrow \text{Upper fence} = Q_3 + 1.5 \times IQR = 27 + 12.0 = 39.0.$$

$$\rightarrow \text{Lower fence} = Q_1 - 1.5 \times IQR = 19 - 12.0 = 7.0.$$

$\rightarrow 38$ is within $[7.0, 39.0]$ $\rightarrow 38$ is NOT an outlier.

Answer: Fences: $[7.0, 39.0]$; 38 is NOT an outlier



Range

22. Find the range:

$\{1, 17, 42, 28, 50\}$

→ $Range = max - min$.

→ $max = 50, min = 1$.

→ $Range = \{max_val\} - \{min_val\} = \{answer\}$.

Answer: Range = 49

23. Find the range:

$\{23, 33, 13, 5, 49, 4\}$

→ $Range = max - min$.

→ $max = 49, min = 4$.

→ $Range = \{max_val\} - \{min_val\} = \{answer\}$.

Answer: Range = 45

24. Find the range:

$\{18, 16, 15, 49, 20\}$

→ $Range = max - min$.

→ $max = 49, min = 15$.

→ $Range = \{max_val\} - \{min_val\} = \{answer\}$.

Answer: Range = 34



Sample variance and standard deviation

25. Find the sample variance and standard deviation (round to 2 decimal places):

{11, 6, 8, 12, 2, 4, 18}

→ Mean = 8.7143.

→ Squared deviations: $(11-8.71)^2=5.22$, $(6-8.71)^2=7.37$, $(8-8.71)^2=0.51$, $(12-8.71)^2=10.8$, $(2-8.71)^2=45.08$, $(4-8.71)^2=22.22$, $(18-8.71)^2=86.22$.

→ $s^2 = \text{sum} \div (n-1) = 177.4286 \div 6 = 29.57$.

→ $s = \sqrt{29.57} \approx 5.44$.

Answer: $s^2 = 29.57$, $s \approx 5.44$

26. Find the sample variance and standard deviation (round to 2 decimal places):

{16, 8, 19, 13, 20, 12}

→ Mean = 14.6667.

→ Squared deviations: $(16-14.67)^2=1.78$, $(8-14.67)^2=44.44$, $(19-14.67)^2=18.78$, $(13-14.67)^2=2.78$, $(20-14.67)^2=28.44$, $(12-14.67)^2=7.11$.

→ $s^2 = \text{sum} \div (n-1) = 103.3333 \div 5 = 20.67$.

→ $s = \sqrt{20.67} \approx 4.55$.

Answer: $s^2 = 20.67$, $s \approx 4.55$

27. Find the sample variance and standard deviation (round to 2 decimal places):

{17, 2, 14, 18}

→ Mean = 12.75.

→ Squared deviations: $(17-12.75)^2=18.06$, $(2-12.75)^2=115.56$, $(14-12.75)^2=1.56$, $(18-12.75)^2=27.56$.

→ $s^2 = \text{sum} \div (n-1) = 162.75 \div 3 = 54.25$.

→ $s = \sqrt{54.25} \approx 7.37$.

Answer: $s^2 = 54.25$, $s \approx 7.37$



Z-score

28. Find the z-score and interpret it:

$$x = 80, \quad \mu = 65, \quad \sigma = 11$$

$$\rightarrow z = (x - \mu) / \sigma.$$

$$\rightarrow z = (\{x_val\} - \{\mu\}) / \{\sigma\} = \{diff\} / \{\sigma\} = \{answer\}.$$

$\rightarrow z = 1.36$: score is 1.36 standard deviation(s) above the mean.

Answer: $z = \frac{80 - 65}{11} = 1.36$

29. Find the z-score and interpret it:

$$x = 96, \quad \mu = 81, \quad \sigma = 5$$

$$\rightarrow z = (x - \mu) / \sigma.$$

$$\rightarrow z = (\{x_val\} - \{\mu\}) / \{\sigma\} = \{diff\} / \{\sigma\} = \{answer\}.$$

$\rightarrow z = 3.0$: score is 3.0 standard deviation(s) above the mean.

Answer: $z = \frac{96 - 81}{5} = 3.0$

30. Find the z-score and interpret it:

$$x = 89, \quad \mu = 70, \quad \sigma = 11$$

$$\rightarrow z = (x - \mu) / \sigma.$$

$$\rightarrow z = (\{x_val\} - \{\mu\}) / \{\sigma\} = \{diff\} / \{\sigma\} = \{answer\}.$$

$\rightarrow z = 1.73$: score is 1.73 standard deviation(s) above the mean.

Answer: $z = \frac{89 - 70}{11} = 1.73$

